Surya Teja Padakanti

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SUMMARY

Seeking data engineering opportunities where I can leverage my experience in ETL, SQL, Python, and cloud technologies to develop and optimize data-driven solutions.

EXPERIENCE

Data Engineering Intern | *IN BOLD PRINT, Remote, United States* **May 2024 – Present**

- **Third-Party API Integration:** Automated emissions factor calculations by integrating a third-party API with secure authentication and parameter configurations, reducing manual effort by 85%.
- **Data Analysis and Field Mapping:** Analyzed transaction data to identify and map essential fields, ensuring accurate input configurations for emissions calculations and enhancing API response accuracy.
- **ETL & Snowflake Integration:** Developed an ETL process using Python and Snowflake connectors to streamline data extraction, transformation, and loading in to database while ensuring consistent data type mapping.
- **Automated Workflow:** Created a **Google Cloud Function** to trigger emissions calculations via HTTP requests, supporting automated, scalable workflows for real-time processing.
- **Quality Assurance and Data Validation:** Established QA checks for emissions data accuracy, including consistency verification and inflation adjustments based on transaction and emissions factor years.
- **Batch Processing Optimization:** Optimized code to handle batch data processing, achieving a 25% reduction in processing time by enhancing system output and memory efficiency.

Data Science Teaching Assistant | *Data Science Department, Clarkson University, Potsdam, United State* **Oct 2023 – Present**

- Extracted data from various sources and integrated it into the university's MySQL Server to enhance database teaching options. Developed SQL questions and queries based on the extracted data to support the teaching.
- Served as a Teaching Assistant for Data Warehouse and database courses, guiding students in designing relational and star schemas, implementing ETL processes, and performing data extraction for analysis. Managed a 30% increase in course enrollment, ensuring all students remained up-to-date and improving overall course performance, resulting in a 10% increase in student achievement compared to previous years.

Research & development, Co-founder | *Drones Origin PVT LTD, Hyderabad, India* **Mar 2021 – Jun 2023**

- Led successful drone workshops where each team constructed a stable, functional drone.
- Developed a thrust stand equipped with a load cell, Arduino, and sensors to measure voltage, current, RPM, thrust, and torque. This data was utilized to optimize the performance of a 1045-propeller.
- Engineered a propeller with 10 distinct airfoil profiles distributed across the half-span of the propeller, leading to a 35% improvement in efficiency. The propeller's performance was validated through experimental testing using the thrust stand and compared with theoretical and computational results.
- Generated theoretical data using airfoil aerodynamic theories to analyze propeller acoustics, assessing sound generation based on factors like camber line, maximum thickness, and maximum chord. This work led to more efficient propeller designs, reducing noise by 5%.

Drone development and Machine learning intern | *JobXRobot Pvt Ltd, Hyderabad, India* **Sep 2021 – Feb 2022**

- Developed a stable drone that met the organization's automation requirements.
- Coordinated with Drone and IT teams to develop algorithms using real-time images of the surroundings and applied image processing techniques with TensorFlow to automate tasks such as navigation, object detection, and obstacle avoidance.

SKILLS & CERTIFICATIONS

- **Languages:** Python, R, SQL, SciPy, TensorFlow, Scikit-learn
- **Databases:** MySQL, Snowflake, IBM Db2
- **ETL Tools:** Snowflake Connector, pymysql, dbt
- **Cloud Platforms:** Microsoft Azure, Google Cloud Platform
- **Data Visualization Tools:** Seaborn, Matplotlib, Tableau, Power BI
- **Productivity Tools:** Microsoft Office (Word, Excel, PowerPoint)
- **Soft Skills:** Communication, Teamwork, Project Management
- **Certifications:** IBM Data Engineering (Coursera), Google Cloud Database engineer (Coursera)

EDUCATION

Ms in Applied Data Science | *Clarkson University, Potsdam, New York, United States* ***Aug 2023 – Dec 2024***

Relevant Coursework: Database Model, Design and Implementation, Data Warehousing, Information Visualization, Data Mining, Machine Learning, Big Data.

CGPA: 4.0/4.0

ACADEMIC PROJECTS

Road Traffic Vs Air Traffic ***Jan 2024 – May 2024***

- Investigated correlations between road and air traffic in NYC to enhance transportation decisions.
- Managed 60GB of air traffic data by, automating extraction and processing of zip files into single rows of essential data using Python modules (zipfile, os, Paramiko, csv, and json), reducing data transformation time from several days to just 140 minutes
- Scaled road traffic data using the min-max technique to address limitations in observations compared to air traffic data, facilitating a more comprehensive analysis.

Forecasting Diabetes ***Jan 2024 – May 2024***

- Developed a machine learning model in Python to predict diabetes risk using the Diabetes Prediction dataset from Kaggle.
- Utilized medical and demographic factors (BMI, hypertension, heart disease, blood glucose) and employed ensemble methods to create 11 balanced subsets, addressing data imbalance with random forest and decision tree algorithms.
- Conducted variable analysis using Seaborn, which demonstrated a strong correlation between BMI and health conditions, confirming the model's findings.

COVID-19 India Dashboard Analysis ***Sep 2023 – Dec 2023***

- Analyzed COVID-19 data for India, focusing on trends in cases, recoveries, and deaths.
- Created interactive dashboards in Tableau to show the spread and effects of the pandemic across states of India.
- Used statistical methods in R to identify patterns and relationships between the variables in the data
- Presented findings with dynamic visuals, making the data easy to understand for decision-makers.